BRYAN JONATHAN

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Recent Mathematics graduate from Universitas Indonesia and BNSP-Certified Data Scientist with hands-on Python experience in machine learning, computer vision, and retrieval-augmented generation. Developed and evaluated customer classification models, researched ConvNeXt-based image classification for an undergraduate thesis, and built a Flask RAG chatbot with PDF ingestion and Chroma vector search. During a six-month internship at Astra Credit Companies, automated data validation with Python, reducing processing time from 15 minutes to 2 minutes.

EDUCATION

Universitas Indonesia Jun 2022 – Aug 2026
Bachelor of Mathematics | GPA: 3.30/4.00

SKILLS

- **Programming & Tools**: Python, R, SQL (SQLite, MySQL), Git/GitHub, Excel, Power BI, JavaScript, HTML, CSS
- **ML & Data Libraries**: PyTorch, torchvision, TensorFlow/Keras, scikit-learn, pandas, NumPy, Matplotlib, Seaborn
- **GenAI & Web**: LangChain, ChromaDB, Sentence Transformers, Gemini API, Flask
- **Methods**: Classification, CNNs, LSTMs, RAG, vector search, model evaluation, statistical and spatial analysis
- **Languages**: Indonesian, English (TOEFL ITP: 640/677)

WORK EXPERIENCE

Astra Credit Companies Mar 2026 – Sep 2026
Data Intern (Brand Management Retention)

- Automated SOLVE package record validation using Python, reducing processing time from 15 minutes to 2 minutes across multiple data sources
- Produced and validated cumulative daily application-status and weekly F9PD reports in Excel, processing 47,000+ source rows at month-end to analyze application volumes, status rates, and growth.
- Maintained a SharePoint-hosted performance dashboard tracking targets, actual results, carry-over, and product trends across 77 branches in six areas.

Universitas Indonesia Aug 2024 – Jan 2025
Assistant Lecturer of Calculus 3

- Mentored 76 students across 9 Calculus 3 sessions, contributing to a 20% improvement in class average scores on midterm and final exams
- Graded assignments and provided detailed feedback to enhance students' mathematical problem-solving skills
- Designed and distributed 90+ supplementary practice problems and study guides to improve student engagement and understanding of complex topics

PROJECT EXPERIENCE

ConvNeXt–KAN Image Classification (Undergraduate Thesis) [Link] 2026

- Built and compared KAN and MLP classification heads on a frozen, pretrained ConvNeXt-Tiny backbone using PyTorch on CIFAR-10.
- Evaluated both models across 10 runs; ConvNeXt–KAN achieved 94.84% mean final test accuracy versus 94.65% for ConvNeXt–MLP.
- Deployed a Dockerized Flask app on Hugging Face Spaces for image uploads and side-by-side ConvNeXt–KAN and ConvNeXt–MLP predictions with class probabilities.

PDAM Customer Tariff Classification for Audit Prioritization [Link] Aug 2026

- Cleaned and validated 10,000 PDAM Surabaya customer records, addressing anomalies, missing values, and class imbalance across 13 tariff classes
- Engineered customer profile features and compared Logistic Regression and Random Forest using stratified 5-fold cross-validation with Macro F1
- Achieved 99.61% Macro F1 on held-out test data and translated model outputs into tariff audit prioritization recommendations

Veritas: Chatbot Assistant for Universitas Indonesia [Link] Dec 2025

- Built a Python/Flask RAG chatbot that answers Universitas Indonesia questions using Gemini and a curated PDF knowledge base.
- Created a PDF ingestion pipeline with multilingual embeddings and Chroma vector search to retrieve relevant passages.
- Developed a multi-session chat interface with follow-up handling and a prompt-based fallback for unsupported questions.

Poverty Determinants in East Java: Spatial & Machine Learning Analysis [Link] Jun 2025

- Modeled poverty across 38 districts/cities using OLS, Spatial Lag/Error models, and Spatial Random Forest in R.
- Tested spatial dependence using adjacency-based weights, Moran's I, and residual diagnostics.
- Achieved R^2 of 0.687 and RMSE of 1.042 with Spatial Random Forest, identifying average years of schooling as the strongest predictor.

Hardware Store Sales and Order Management System [Link] Jun 2025

- Designed a six-table SQLite database covering customers, products, orders, payments, and administrator accounts.
- Built a role-based Python interface for registration, product search, cart checkout, inventory updates, and order management.
- Implemented parameterized CRUD operations, stock validation, and multi-table SQL joins for transaction processing and reporting.

East Java Poverty Analysis Using Geographically Weighted Regression [Link] Apr 2025

- Combined socioeconomic indicators with spatial data for 38 East Java districts/cities using R.
- Compared OLS with three GWR kernels; Bisquare GWR improved AIC from 166.38 to 157.26 and R^2 from 0.808 to 0.823.
- Visualized local coefficients, statistical significance, and residuals to identify geographically varying poverty relationships.

ORGANIZATION AND COMMITTEE EXPERIENCE

OIM MATEK 2023 Jul 2023 – Dec 2023

Berkas Staff

- Contacted and scouted alumni to work as judges for poster competition.
- Worked as Liaison Officer, managing communication and coordination with judges during the event.
- Served as the main point of contact for poster competition participants, ensuring an efficient event experience.

ONLINE COURSES AND CERTIFICATIONS

- Certified Data Scientist — BNSP [Link] Oct 2026 - Oct 2029
- TOEFL ITP: 640/677 — ETS [Link] Jul 2026 - Jul 2028